Communication lower bounds for Multi-Party Graph Traversal

Ryan Batubara, Ivy Hawks, Darren Ho, Ciro Zhang

Mentor: Shachar Lovett, *slovett@ucsd.edu*

Abstract

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Motivation

- Figure of multiple agents on graph
- Point about good algorithm designers finding $\mathcal{O}$ bounds
- CC allows for finding Ω bounds, reducing problem scope for algorithm designers

Basic Example: Equality

There are two players, traditionally called **Alice** and **Bob**.

- Alice holds input x
- Bob holds input y
- Together, they want to compute a function $f(x, y)$

They are allowed to send messages (bits) back and forth. The goal is to minimize the total number of bits communicated.

Equality wishes to compute if two bit-strings are the same:

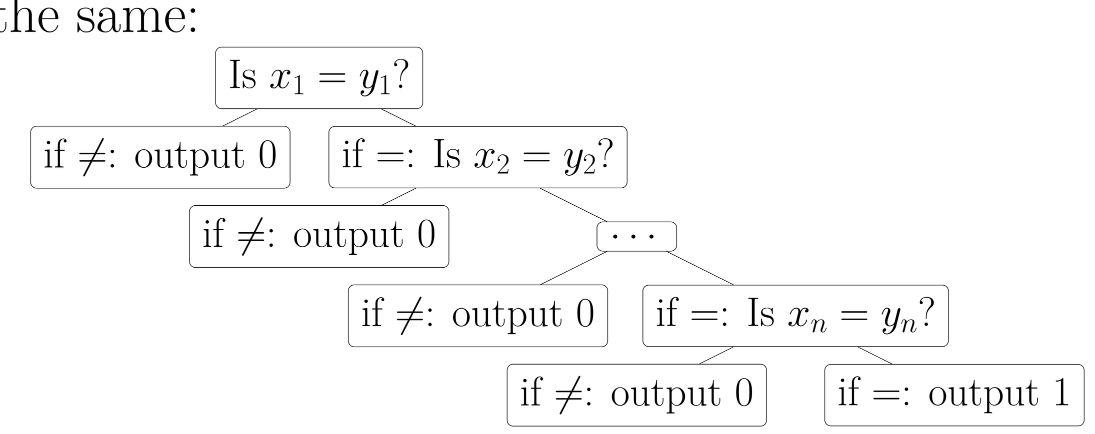

Figure 1:Visual for Equality Proof

Figure 3 demonstrates how, in the worst case, we must check every bit, since if every bit except x_n and y_n are the same, we will have to check all n bits to show equality. Thus, the deterministic communication complexity of Equality is $D(\text{EQ}) = n$

Two Player TRAV with EQ

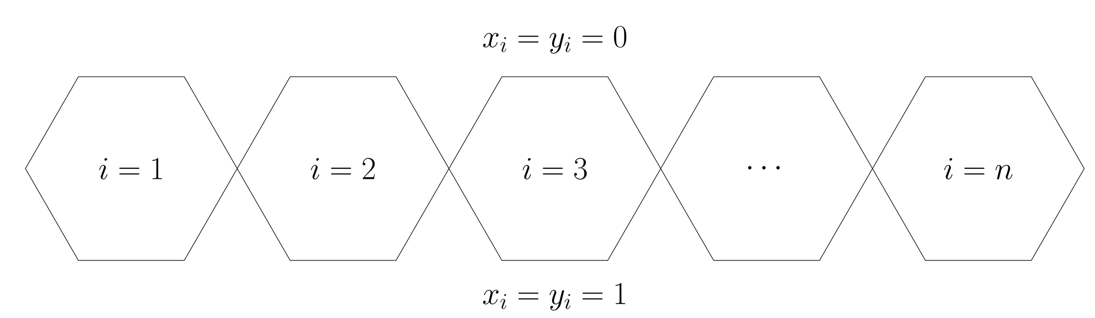

Figure 2:equality gadgets

1. The top branch corresponds to $x_i = y_i = 1$,
2. The bottom branch corresponds to $x_i = y_i = 0$.

Multi-Player TRAV, disjoint graphs

Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin. Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin.

$$E = mc^2 \tag{1}$$

Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin. Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin.

$$\cos^3 \theta = \frac{1}{4} \cos \theta + \frac{3}{4} \cos 3\theta \tag{2}$$

Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin. Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin.

Two Player TRAV with DISJ.

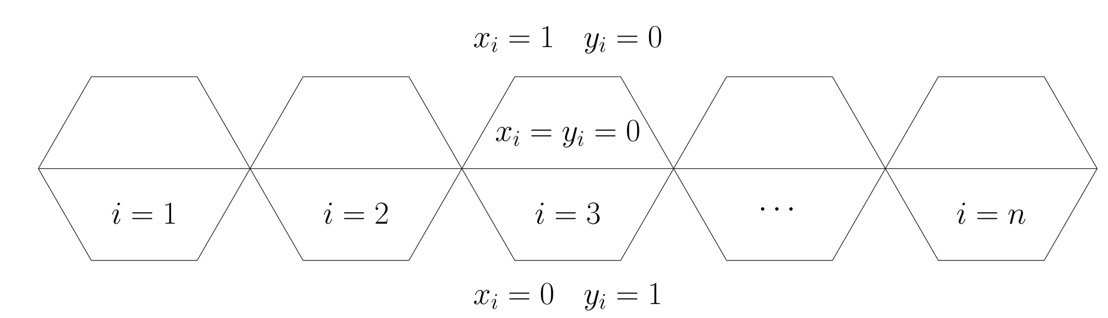

Figure 3:disjoint gadgets

1. The top branch corresponds to when x_i has the element but y_i does not
2. The middle branch corresponds to when x_i and y_i do not have the element
3. The top branch corresponds to when y_i has the element but x_i does not

Multi-Player TRAV, connected graphs

Figure 4:Figure caption

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Two-player TRAV, iterative

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Multi-Player TRAV, iterative

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- Curabitur pellentesque dignissim
- Eu facilisis est tempus quis
- Duis porta consequat lorem

References

Acknowledgements

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